A label-free electrochemical biosensor for the detection of alpha-thalassemia 1 (SEA deletion) carriers using screen-printed carbon electrodes

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ABSTRACT

A label-free electrochemical DNA biosensor based on screen-printed carbon electrodes (SPCEs) coupled with H33258 dye has been extensively developed to diagnose human genetic diseases. However, many developed methods are still limited to simulated target DNA. In the present study we aimed to develop a label-free electrochemical DNA biosensor based on SPCEs for identifying carriers of α-thalassemia 1 (SEA deletion), which is the most common inherited blood disorder worldwide. Twenty samples of known normal α-globin gene and α-thalassemia 1 (SEA deletion) carriers were assessed; the criterion cut-off values were more than 26.84 kΩ (negative imaginary impedance, -Z") and 48.83 kΩ (charge transfer resistance, Rct). Reliability analysis between -Z" and Rct showed a strong correlation, with a Pearson's correlation coefficient value of 0.88 and a mean difference between Rct and -Z″, determined by Bland–Altman plot analysis, of 55.8 kΩ. The diagnostic potency of the developed EIS biosensor based on SPCEs was evaluated in 81 clinical blood samples. The results demonstrate that the biosensor had 100% sensitivity, 88.2% specificity, 83.3% positive predictive value (PPV), 100% negative predictive value (NPV) and 92.6% accuracy for Rct measurement. The developed EIS biosensor based on SPCEs is an attractive alternative method to gel electrophoresis for the detection of α-thalassemia 1 (SEA deletion) carriers. It is simple, inexpensive and can reduce turnaround time 45-fold. Identification of an α-thalassemia 1 (SEA deletion) carrier is the most effective approach for the management of thalassemia and for the prevention of Hb Bart's hydrops fetalis, the most severe form of thalassemia.

Keywords:

Alpha-thalassemia 1 carrier, Diagnostic performance, DNA biosensor, Electrochemical biosensor, Label-free DNA detection, Molecular diagnosis

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1. Introduction

The creation of a rapid, simple, affordable, portable, highly sensitive and specific method of DNA detection for use in genetic disease diagnosis and prenatal genetic counselling has been the focus of much recent attention, and various DNA detection platforms have been developed in order to achieve this. Label-free DNA biosensors have advanced significantly in recent years due to their capacity for rapid response, easy operation and high sensitivity. A variety of label-free DNA biosensors have been proposed for the detection of DNA sequences using optical, piezoelectric and electrochemical techniques (Anusha et al., 2022; Mehrotra, 2016). Among these technologies, electrochemical biosensors have prominent advantages, such as real-time analysis and cost-effectiveness. Screen-printed carbon electrodes (SPCEs) exhibit favorable properties, such as low cost, high chemical stability, relative inert electrochemistry, rich surface chemistry, and electrocatalytic activity for a variety of redox reactions (Moreira et al., 2016). Therefore, they have been extensively used in electrochemical biosensors for various applications, such as the detection of foodborne pathogens, infectious diseases, chemical substances in cosmetic products and abnormal genes. Electrochemical impedance spectroscopy (EIS), which is simple, easy to use, portable, cost effective and disposable, is one of the most important electrochemical techniques (Hayat and Marty, 2014; Monteil et al., 2021; Ozcelikay et al., 2019). All these characteristics make EIS ideal for point-of-care devices. A label-free DNA biosensor using EIS based on SPCEs operates on the principle that redox molecules that are usually not highly electroactive will generate an electrochemical signal through a carbon electrode after binding to double-stranded DNA (ds-DNA). Hoechst 33258, a popular fluorescent stain, is a particularly effective redox probe for DNA detection. When ds-DNA is present, Hoechst 33258 will bind and quench the electro-oxidation, causing a reduction in

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electrochemical current. The electrochemical responses (the changes in current and potential) of SPCEs coupled with Hoechst 33258 depend on the amount of ds-DNA (Ahmed et al., 2007; Hughes et al., 2016). This technique was developed to determine the Salmonella and Hepatitis B virus gene amplified by PCR, and the pathogenic *E. coli* O157 gene amplified by LAMP (Jaroenram et al., 2020; Kampeera et al., 2019). Most of the developed SPCEs coupled with Hoechst 33258 in EIS are used to identify bacterial, viral or pathogenic genes. The development of a method for the detection of human genes has been limited due to their complexity. There are several reports on the development of EIS biosensors for the detection of abnormal human genes, such as the BCR/ABL fusion gene in chronic myelogenous leukemia (CML) (Chen et al., 2008), the IVSII-1 gene in β-thalassemia (Gholivand and Akbari, 2019), and the HLA-B*15:02 and HLA-B*15:21 genes in carbamazepine-induced Stevens–Johnson syndrome (SJS) or toxic epidermal necrolysis (TEN) (Chomean et al., 2022).

The detection of α-thalassemia 1 (SEA deletion or $-^{SEA}/\alpha\alpha$) carriers is an effective way to avoid the birth of patients with severe thalassemia (Chaibunruang et al., 2018; Hockham et al., 2019; Pharephan et al., 2016). The abnormal α-globin gene causing α-thalassemia 1 (SEA deletion) in a carrier result from a large DNA deletion in both α-globin genes spanning approximately 20.5 kb. Couples who are both carriers of this deletion have a 25% chance of having a child affected by Hb Bart's hydrops fetalis (--/--), which is the most severe form of thalassemia, usually leading to death in utero or shortly after birth (Vichinsky, 2013). Therefore, the identification of α-thalassemia 1 (SEA deletion) carriers is included in the prenatal screening and preimplantation genetic testing program for couples who are at high risk of having a baby with Hb Bart's hydrops fetalis. A label-free DNA biosensor based on a quartz crystal microbalance (QCM) was successfully developed to detect an abnormal α-globin gene causing α-

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thalassemia 1 (SEA deletion) (Chomean et al., 2010). However, preparation of the DNA probe immobilized on the electrode surface is required. In the past decade, electrochemical DNA biosensors have received a great deal of interest for use in thalassemia detection. However, most of the developed electrochemical DNA biosensors are only suitable for detecting single-base mismatch or non-deletional α- and β-thalassemia mutations.

In the present study we aimed to develop a label-free DNA electrochemical biosensor based on EIS measurement for identifying α-thalassemia 1 (SEA deletion) carriers. A rapid and sensitive EIS biosensor coupled with multiplex polymerase chain reaction (mPCR) is a critical platform for controlling new-birth cases of severe thalassemia.

2. Material and methods

2.1 Study population

Ethical approval for this study was provided by the Third Ethics Committee of Thammasat University, Pathum Thani Province, Thailand (COA No. 066/2565).

Twenty known genomic DNA samples of each normal individual and α-thalassemia 1 (SEA deletion) carrier were collected from division of Medical Technology Laboratory, Thammasat University Hospital (TUH) to evaluate the optimal EIS measurement conditions for identifying an α-thalassemia 1 (SEA deletion) carrier. The known DNA samples were also confirmed by PCR as previously reported (Chomean et al., 2018). The EIS responses were analyzed by area under the ROC curve in order to discriminate α-thalassemia 1 (SEA deletion) carriers from normal individuals.

The diagnostic ability of the developed EIS measurement based on SPCEs for identifying α-thalassemia 1 (SEA deletion) carriers was tested on 81 clinical EDTA blood samples with

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mean corpuscular volume (MCV) less than 80 femtoliters (fL) from division of Medical Technology Laboratory, Thammasat University Hospital (TUH). Genomic DNA was extracted from all blood samples using a Gentra Puregene Blood Kit (QIAGEN, Germany). Genomic DNA quality and quantity were determined using a NanoDrop One/ OneC Microvolume UV–Vis Spectrophotometer (Thermo Fisher Scientific, USA). The extracted genomic DNA was aliquoted and stored at -20 ◦C until use.

2.2 Fabrication of screen-printed carbon electrodes (SPCEs)

The geometric design of the electrodes was performed using Adobe Illustrator as displayed in Supplemental Data Fig. S1. The design comprised a working electrode (WE) active area of 12.57 mm^2 and a counter electrode (CE) active area of 15.70 mm^2. Each SPCE has 12 mm in width and 25 mm in height. The carbon paste was screen-printed as the first layer of the WE and CE onto a poly-ethylene terephthalate (PET) substrate. The second layer was printed with Ag/AgCl at a ratio of 60/40 for the reference electrode (RE). The substrates were baked at 60°C for 5 min in an oven to remove solvents after printing the first and second layers. The last layer was printed with insulator pastes and dried at 120°C for 5 min.

Before testing, the printed electrodes were cleaned with 0.1M PBS solution (pH 7.4) to remove contaminants on surface, by applying a voltage between 0.0 and +1.0 V in cyclic voltammetry (CV) mode for 6 cycles at a scan rate of 100 mV/s. CV was performed using a PalmSens 4 potentiostat and PSTrace 5 software (PalmSens BV, Netherlands).

2.3 Preparation of mPCR product amplified from normal α-globin and α-thalassemia 1 (SEA deletion) gene

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Four sets of primers were designed using Primer 3 software according to sequence accession number AY207443 (Supplemental Data Table 1). Each primer set was used to amplify the normal α-globin gene and abnormal α-globin gene causing α-thalassemia 1 (SEA deletion) by (mPCR). The mPCR was performed in a final volume of 20 µL containing 100 ng of extracted genomic DNA. Each reaction consisted of 10 µL of 2x PCR master mix (GoTaq Master Mix, Promega Corporation, USA) and each concentration of primer, as shown in Supplemental Data Table 1. The amplification of the target gene was carried out for 35 cycles of 95°C for 30 sec, 63°C for 30 sec and 72°C for 30 sec. All mPCR products were verified by 1.5% agarose gel electrophoresis and visualized under UV illumination.

2.4 Evaluation of EIS biosensor based on SPCEs for identifying α-thalassemia 1 (SEA deletion) carriers

EIS was performed on SPCEs using a portable PalmSens4 analyzer. The change in electrochemical signal was used to discriminate the normal from the abnormal α-globin genes, causing α-thalassemia 1 (SEA deletion), after binding to Hoechst 33258. The experimental strategy for the developed EIS measurement based on SPCEs for distinguishing α-thalassemia 1 (SEA deletion) carriers from normal individuals is shown in Fig. 1.

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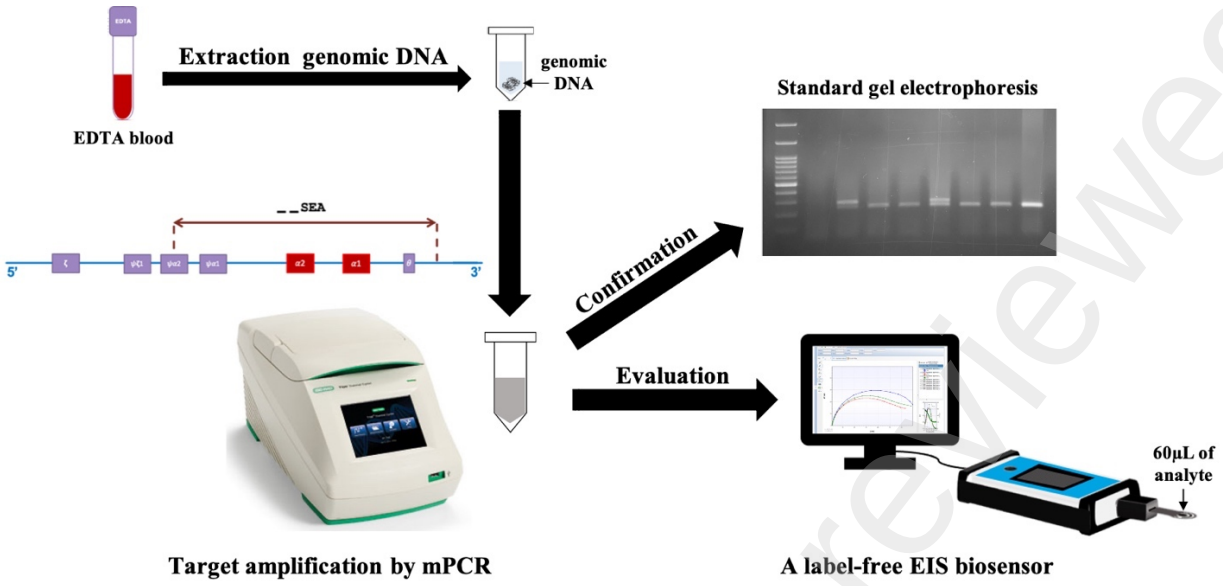

Schematic Fig. 1. Experimental strategy for a label-free EIS biosensor based on SPCEs and Hoechst 33258 for the detection of abnormal α-globin genes causing α-thalassemia 1 (SEA deletion).

To assess the optimal primer sets giving different electrochemical signals in normal individuals and α-thalassemia 1 (SEA deletion) carriers, each mPCR amplicon was analyzed. Six microliters of mPCR amplicon were mixed with 0.01 M Hoechst 33258, in a final volume 60 µL, and dropped onto the electrode surface. The impedance parameter was registered at frequencies ranging from 0.1 Hz to 10 kHz at a constant potential of 300 mV and an alternating potential of 10 mV.

The limit of detection (LOD) was determined based on the lowest concentration of extracted genomic DNA that could distinguish an α-thalassemia 1 (SEA deletion) carrier from a normal individual. A serial 2-fold dilution, generating genomic DNA of 12.5, 25, 50 and 100 ng, was amplified by mPCR using the appropriate primer set. The mPCR amplicons were analyzed using agarose gel electrophoresis and EIS measurement.

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The proper primer set and limit of detection were tested using known samples of DNA containing normal and α-thalassemia 1 α-globin (SEA deletion) in triplicate.

The experiments were carried out using 20 known α-thalassemia 1 (SEA deletion) carrier and normal samples to determine the parameters giving optimal cut-off values. Additionally, distilled water was used as a non-template control (NTC) to monitor contamination and primer dimer formation, which could produce false-positive results.

2.5 Diagnostic performance of the developed EIS biosensor based on SPCEs for identifying α-thalassemia 1 (SEA deletion) carriers

The clinical diagnostic potency of the developed EIS based on SPCEs was evaluated using 81 blood samples of suspected SEA deletion carriers (hypochromic microcytic red blood cells with mean corpuscular volume (MCV) <80 fL). Genomic DNA was extracted and the target DNA was amplified using mPCR. All amplified amplicons were investigated under optimal conditions using both EIS measurement and conventional gel electrophoresis.

2.6 Statistical analysis

Statistical analysis was performed using the MedCalc software (version 18.2.1; Mariakerke, Belgium). Paired t-test were used for the comparative study to determine the difference in electrochemical signals between the groups. The *P* values less than 0.05 indicated a statistically significant difference between the groups with a 95% confidence interval (CI). Method agreement of the EIS responses between the negative imaginary impedance (–Z”) and charge transfer resistance (Rct) was assessed using the Spearman’s rank correlation (ρ) and Bland-Altman analysis. The cut-off value offering suitable sensitivity, specificity, positive likelihood ratio (+LR), negative likelihood ratio (-LR), positive predictive value (PPV), negative

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predictive value (NPV), and accuracy were determined by the developed EIS biosensor comparing to conventional PCR, assuming the reference method.

3. Results

3.1 Preparation of mPCR product amplified from normal α-globin and α-thalassemia 1 (SEA deletion) gene

Because of the reduction in the electrochemical signal due to the binding between ds-DNA and Hoechst 33258, each set of primers was used to amplify normal α-globin and α-thalassemia 1 (SEA deletion) genes. The product sizes obtained from all mPCR amplicons amplified from each set of primers were as expected (Fig. 2A).

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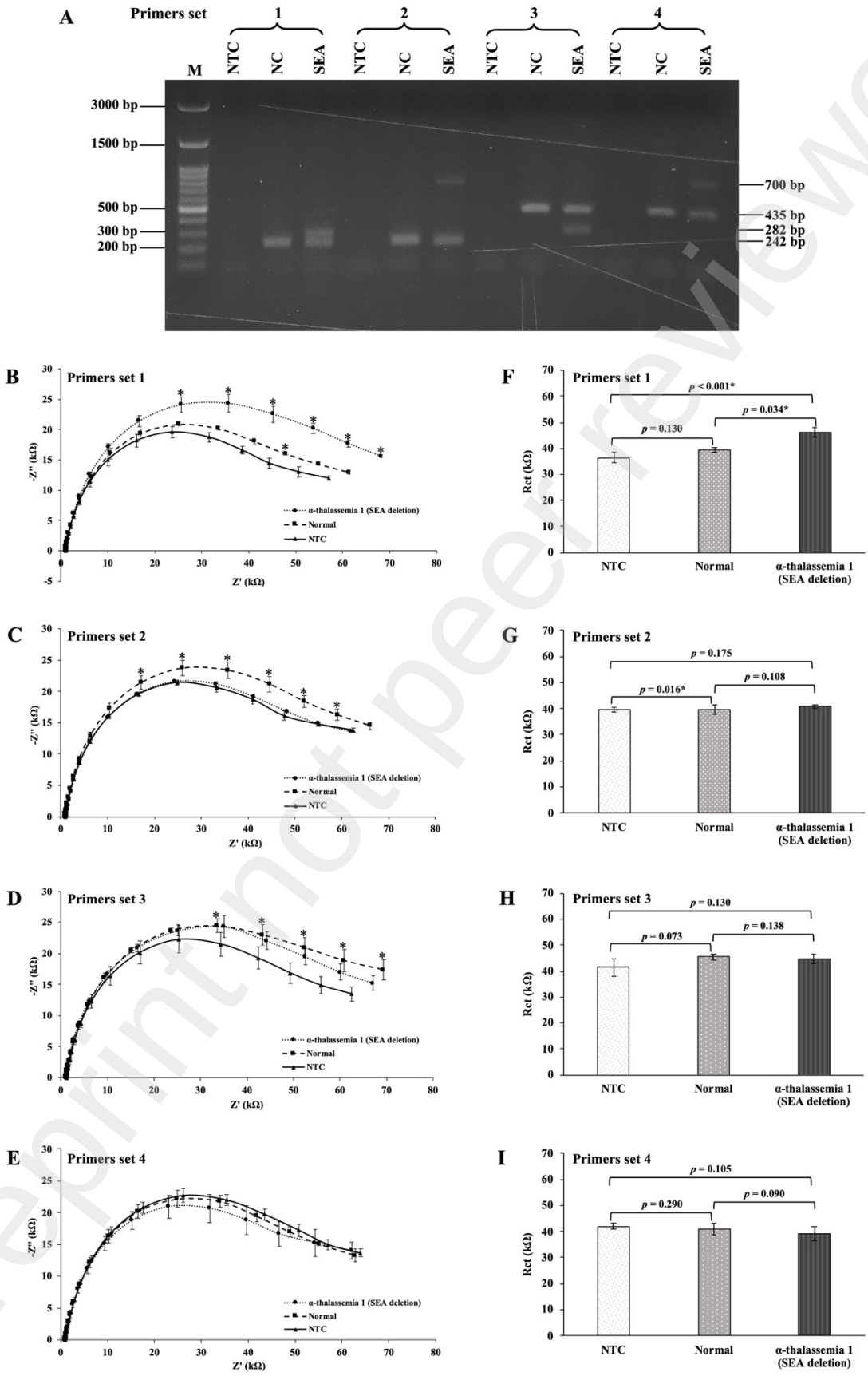

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Fig. 2. Agarose gel electrophoresis of mPCR products (A). No template DNA (NTC, distilled water), normal control (NC, normal α-globin gene) and α-thalassemia 1 with SEA deletion (SEA) were amplified by each set of primers (1–4). M represents the molecular weight markers (3 kb DNA ladder; BioRad). EIS measurement is based on Nyquist plots (B-E) and Rct (F-I) of mPCR products amplified from each primer set (1–4) at frequencies ranging from 0.1 Hz to 10 kHz. The impedance signal was evaluated based on the ability to discriminate an α-thalassemia 1 (SEA deletion) carrier from a normal individual and the non-template control.
*means statistically significant difference ($P<0.05$)

3.2 Evaluation of EIS biosensor based on SPCEs for identifying α-thalassemia 1 (SEA deletion) carriers

To assess the mPCR amplicons allowing the discrimination of electrochemical signals from normal individuals and α-thalassemia 1 (SEA deletion) carriers, the Nyquist plots (–Z", versus Z') and Rct parameters of the EIS biosensor were analyzed. Fig. 2 represents the Nyquist plots (B-E) and Rct (F-I) obtained from the mPCR amplicons amplified by primer sets 1–4, respectively. Primer set 1 showed a statistically significant difference in impedance signals between normal individuals and α-thalassemia 1 (SEA deletion) carriers (Fig. 2B and 2F).

In order to develop a portable mini-potentiostat along with an SPCE, the optimal frequency for identifying α-thalassemia 1 (SEA deletion) carrier was determined. The impedance signals obtained from the mPCR products amplified by primer set 1 were evaluated. Fig. 3 shows that the best discrimination between α-thalassemia 1 (SEA deletion) carrier, a normal individual and the non-template control was at a frequency of 0.58 Hz. Therefore, a frequency of 0.58 Hz was used for EIS measurement throughout the experiments.

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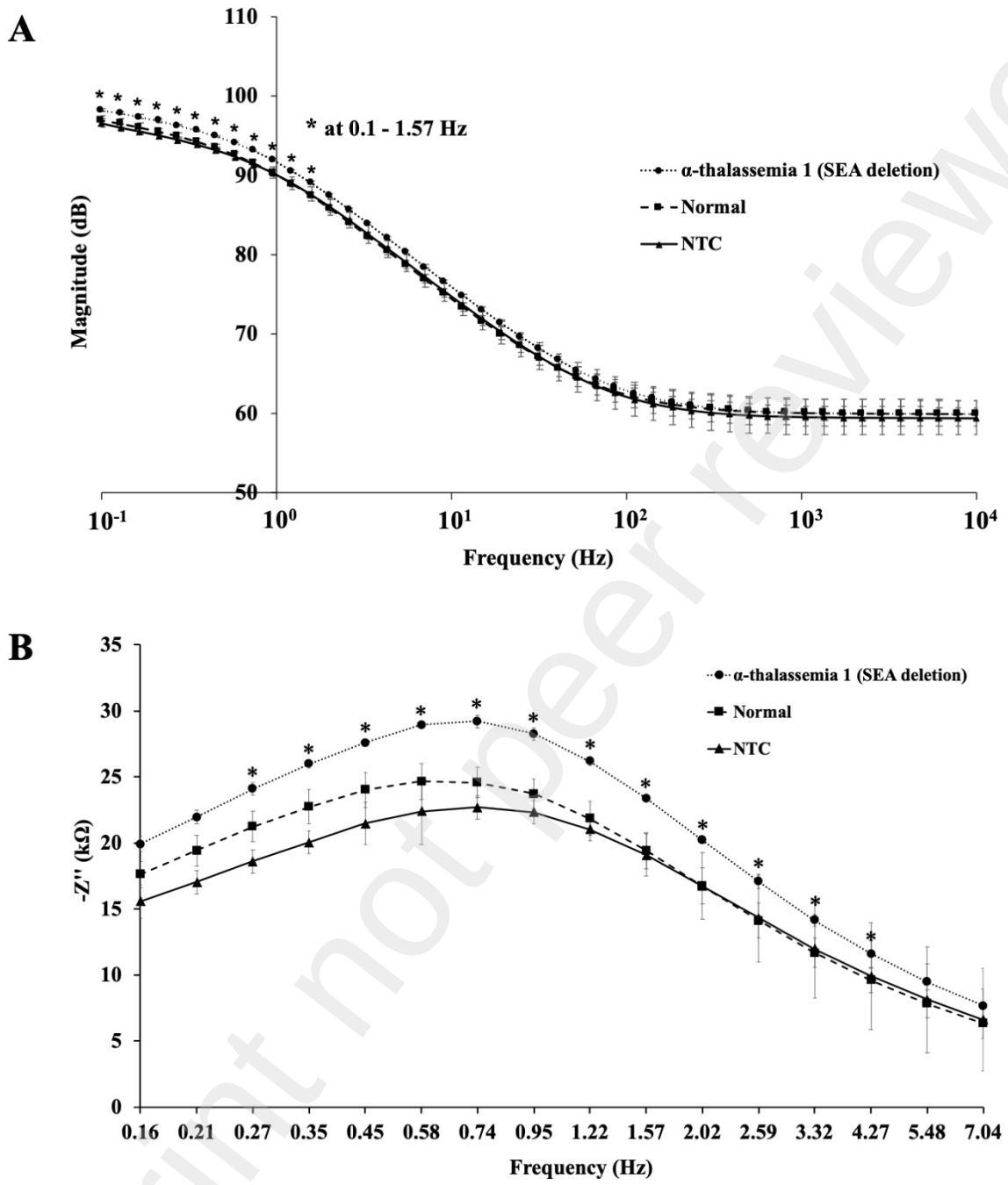

Fig. 3. Bode plot comparing log of frequencies with magnitude (A) and Nyquist plot comparing frequencies with –Z" (B), at 0.16 Hz to 7.04 Hz, of mPCR products amplified from α-thalassemia 1 (SEA deletion) carrier, normal individual and non-template control (NTC).

*means statistically significant difference (P<0.05)

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The sensitivity of the EIS measurement based on SPCEs and H33258 for identifying an α-thalassemia 1 (SEA deletion) carrier was assessed. The -Z" and Rct values increased as DNA concentration increased (Fig. 4). At 12.5 ng of genomic DNA, -Z" and Rct showed a statistical difference between a normal individual and an α-thalassemia 1 (SEA deletion) carrier (Fig. 4A and 4B), respectively. The impedance signals could differentially diagnose an α-thalassemia 1 (SEA deletion) carrier from a normal individual. However, the normal band was not observed when determined by gel electrophoresis (Fig. 4C). Thus, it can be concluded that EIS measurement based on SPCEs and H33258 is two times more sensitive than conventional gel electrophoresis.

The experiments were carried out using 20 known samples from normal individuals and α-thalassemia 1 (SEA deletion) carriers to find the optimal cut-off value at a frequency 0.58 Hz. The means ± SD of Z″ were 25.30±0.54, 25.97±0.87 and 29.98±1.21 kΩ in the non-template control, normal samples and α-thalassemia 1 (SEA deletion) carriers, respectively (Fig. 5A). In addition, the means ± SD of Rct were 45.28±1.64, 46.63±1.08 and 52.32±1.81 kΩ in the NTC, normal samples and α-thalassemia 1 (SEA deletion) carriers, respectively (Fig. 5B). Analysis of the ROC curve demonstrated that the optimal criteria for discriminating α-thalassemia 1 (SEA deletion) carrier from a normal individual were 26.84 and 48.83 kΩ for -Z″ and Rct, respectively. These cut-point values provided 100% sensitivity (95% CI: 83.2 to 100.0) and 100% specificity (95% CI: 83.2 to 100.0).

Correlation analysis for EIS measurement between -Z″ and Rct showed a very strong correlation, with a Pearson's correlation coefficient value of 0.884 (95%CI: 0.799 to 0.934, $P<0.0001$) (Fig. 5C). The mean difference between the Rct and -Z″ was 55.8 kΩ, with a

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95% limit of agreement ranging from 54.9 to 56.6 kΩ, (Fig. 5D). The results show that the majority of EIS measurements were within the 95% limit of agreement with only one outlier.

Finally, the impedance parameters for the development of a mini-potentiostat based on SPCEs for identifying α-thalassemia 1 (SEA deletion) carriers were set at a frequency of 0.58 Hz, a constant potential of 300 mV and an alternate potential of 10 mV. The circuits for α-thalassemia 1 (SEA deletion) carrier, normal individual and non-template control are shown in Supplemental Data Fig. S2.

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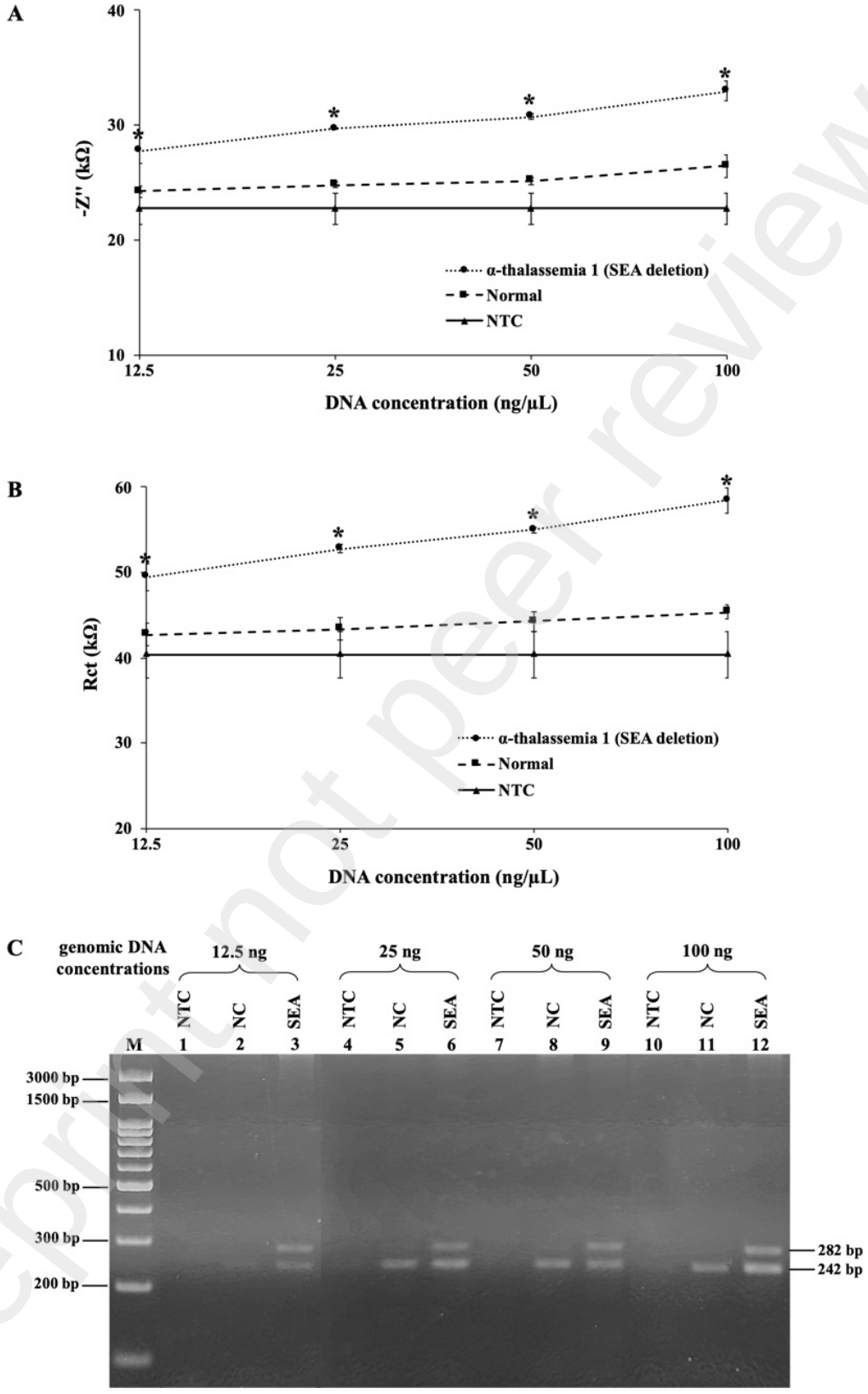

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Fig. 4. Limit of detection of EIS measurement, based on SPCEs and H33258 at 0.58 Hz determined by -Z" (A) and Rct (B), compared to conventional gel electrophoresis (C) for identifying α-thalassemia 1 (SEA deletion) carriers. Genomic DNA of normal individual (NC), and α-thalassemia 1 with SEA deletion (SEA) varied at 2-fold dilution (lanes 1–12). NTC is the control without genomic DNA and M represents the molecular weight markers (3 kb DNA ladder; BioRad).

*means statistically significant difference (P<0.05)

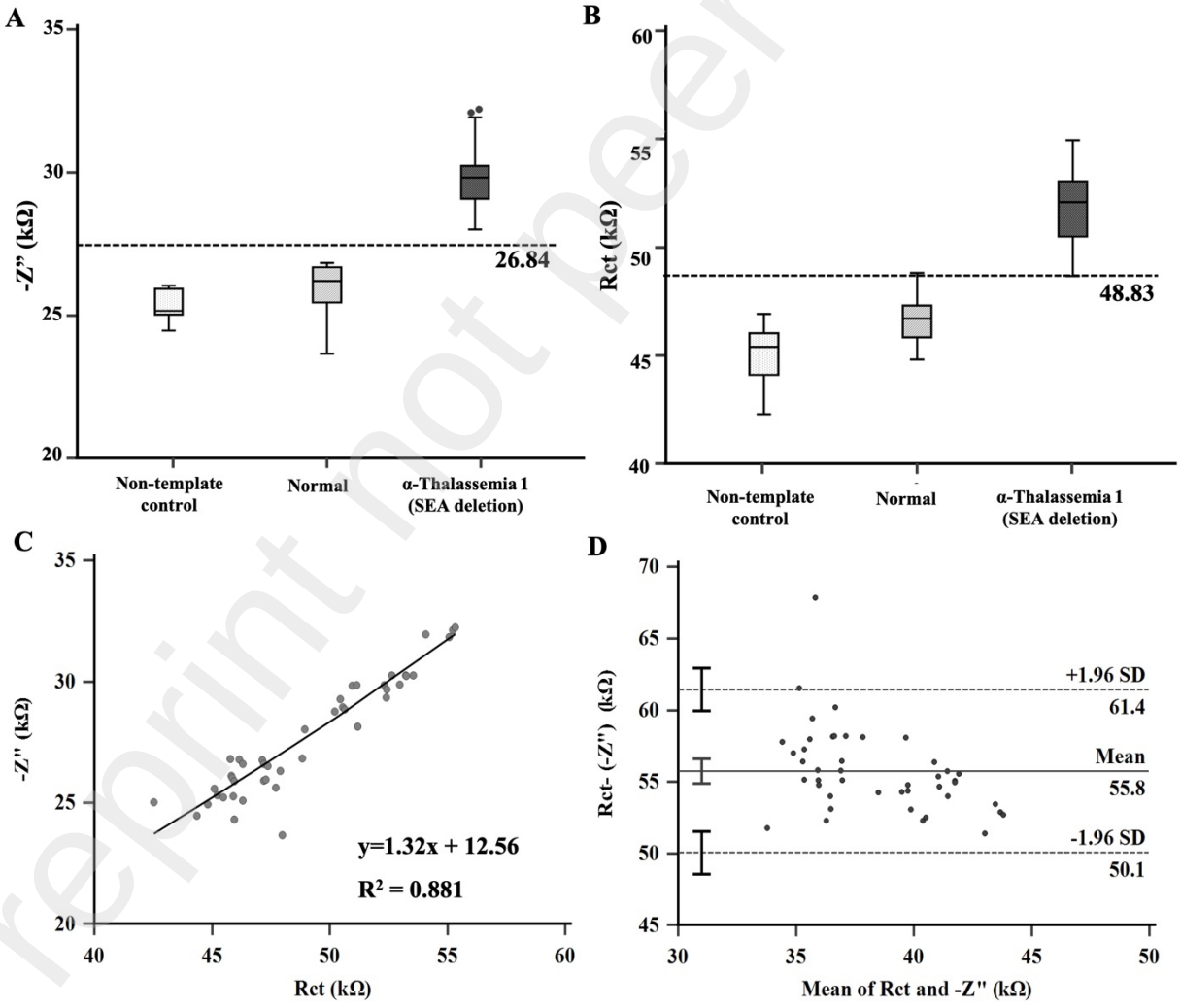

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Fig. 5. Analysis of EIS measurement of the negative imaginary impedance (-Z″; A) and charge transfer resistance (Rct; B) in 20 known normal individual and α-thalassemia 1 (SEA deletion)-carrier samples. Pearson's correlation analysis (C) and Bland–Altman plot (D) between negative imaginary impedance (-Z″) and charge transfer resistance (Rct).

3.3 Diagnostic performance of the developed EIS biosensor based on SPCEs for identifying α-thalassemia 1 (SEA deletion) carriers

All mPCR amplicons were analyzed using the developed EIS biosensor and standard gel electrophoresis. Cut-off values for α-thalassemia 1 (SEA deletion) carrier and normal α-globin-individuals could be determined from either the -Z" or the Rct responses of the known samples. The results reveal that a -Z" value of more than 26.84 kΩ (or an Rct value of more than 48.83 kΩ) can be used to detect α-thalassemia 1 (SEA deletion) carriers. EIS measurement demonstrated false-positive results in 9 and 6 clinical samples analyzed using -Z" and Rct, respectively. The label-free EIS biosensor based on SPCEs demonstrated 100% sensitivity comparable to conventional gel electrophoresis (Table 1). However, the specificity of the developed EIS biosensor based on SPCEs determined by -Z" and Rct was 82.4% and 88.2%, respectively. These findings suggest that the Rct possessed higher specificity and accuracy than the -Z".

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Table 1. Clinical performance of EIS biosensor based on SPCEs for distinguishing α-thalassemia 1 (SEA deletion) carriers from normal individuals.

EIS measurement	Sensitivity (95%CI)	Specificity (95%CI)	Positive likelihood ratio (95% CI)	Negative likelihood ratio (95% CI)	Positive predictive value (95% CI)	Negative predictive value (95% CI)	Accuracy (95% CI)
-Z”	100% (88.4 to 100.0)	82.4% (69.1 to 91.6)	5.7 (3.1 to 10.3)	0	76.9% (64.8 to 85.8)	100%	88.9% (80.0 to 94.8)
Rct	100% (88.4 to 100.0)	88.2% (76.1 to 95.6)	8.5 (4.0 to 18.0)	0	83.3% (70.2 to 91.4)	100%	92.6% (84.6 to 97.2)

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4. Discussion

The current worldwide field of genetic disease diagnosis and personalized medicine has encouraged the development of a rapid, simple, inexpensive, and portable genetic testing method (Kamsong et al., 2022). Because of the complexity of the human genome, the developed electrochemical DNA biosensors therefore limit testing to only synthetic target DNA or simulated samples (Gholivand and Akbari, 2019; Li et al., 2021; Wei et al., 2009; Wu et al., 2020). It has proved challenging for researchers to develop an electrochemical DNA biosensor that can be used for clinical samples. In this study, EIS was used to measure the -Z" and Rct resulting from free H33258 in mPCR mixtures. The responses were directly proportional to the target DNA amplified via mPCR. A few studies have reported that the length of the PCR product and GC content affect EIS measurements (Ahmed et al., 2007; Ahuja et al., 2022; Ikeuchi et al., 2016; Mills et al., 2018). Fig. 2 demonstrates that shorter PCR products (242 bp and 282 bp) gave a higher signal compared to 435 bp and 700 bp PCR products. Similarly, previous work has successfully detected PCR amplicons of SARS-CoV-2 at lengths of 117 bp and 503 bp (Ahuja et al., 2022). The size of the mPCR product significantly affects the sensitivity of EIS, since larger fragments bound to the electrode surface should prevent the access of redox couples (Ahuja et al., 2022; Mills et al., 2018). In addition, increasing the -Z" and Rct signal corresponds to an increase in DNA concentration (Fig. 4). H33258 binds in the minor grooves of AT-rich regions of double-stranded DNA (dsDNA), causing a decrease in current and an increase in resistance of the SPCE surface (Ahmed et al., 2007; Ahuja et al., 2022; Ikeuchi et al., 2016; Mills et al., 2018). Because the binding of H33258 to dsDNA is non-specific, the specificity of the EIS response heavily relies upon the characteristics of the primer and the purity of the mPCR amplification (Ahmed et al., 2007; Ahuja et al., 2022; Ikeuchi et al., 2016; Mills et al., 2018). All

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the primers designed in this study have 35–70% GC content, ranging from 39.13 to 68.42%, and an absence of primer secondary structure (Supplemental Data Table 1). The EIS values for identification of α-thalassemia 1 (SEA deletion) carriers were more than 26.84 kΩ for –Z" and 48.83 kΩ for Rct with 100% sensitivity and specificity, and the area under the ROC curve was 1.0 (Fig. 5). However, the diagnostic potency of the developed EIS biosensor based on SPCEs using Rct measurement was 100% sensitivity and 88.2% specificity (Table 1). These results may be caused by random errors from the presence of inhibitors in clinical blood samples. In addition, EIS measurement by the developed EIS biosensor based on SPCEs could not differentiate between NTC and a normal individual. Therefore, the process of sample collection and genomic DNA extraction should be undertaken with care. The hybridization between the immobilized DNA probe on the SPCE surface and the amplified target DNA is an interesting method for enhancement of the specificity of the developed EIS biosensor (Ahmed et al., 2007; Ahuja et al., 2022; Flauzino et al., 2021; Gholivand and Akbari, 2019; Ikeuchi et al., 2016; Mills et al., 2018; Wei et al., 2009). Detection of an α-thalassemia 1 (SEA deletion) carrier using the developed EIS biosensor based on SPCEs costs $2, which is 20 times lower than conventional methods (PCR and real-time PCR). However, development of the EIS biosensor based on SPCEs coupled with a hybridization method will increase the cost.

5. Conclusions

The development of a label-free EIS biosensor with high sensitivity and specificity based on SPCEs and H33258 for identifying α-thalassemia 1 (SEA deletion) carriers was achieved in this study. The developed method is simple, inexpensive, and requires 1.5 hr. to amplify the target DNA by mPCR and 1 min for EIS measurement. In the future, in order to enhance the field application of EIS biosensors for the identification of α-thalassemia 1 (SEA deletion)

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carriers, especially in remote areas, one-temperature gene amplification, such as loop-mediated isothermal amplification (LAMP) or recombinase polymerase amplification (RPA), could possibly be used.

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